

# Getting Started with Python

If at any point, your output doesn't match what's listed below, please refer to the [Troubleshooting](#) section.

## Requirements

---

- Python 3.5+
- Knowledge of your Moku's IP Address (see below)

## Quickstart

---

Download the mokuccli installer from [Utilities](#)  and follow the installer prompts to install the mokuccli commandline tool. Then download the required firmware version as below, you can easily check that the installation succeeded by running the simple Python command listed below.

```
1    $ pip install moku
2    $ mokuccli download 600
3    $ python
4    >>> from moku.instruments import Oscilloscope
5    >>> osc = Oscilloscope('192.168.###.###')
6    >>> osc.get_data()
```

You should see an array of data captured from your newly-deployed Oscilloscope object. If not, please follow the [Full Procedure](#) below or refer to the [Troubleshooting](#) section.

## Full Procedure

---

### 1. Check your Python Installation

(e.g. 2.7.0 ) is not supported.

```
1 $: python --version
2 Python 3.9.0
```

## 2. Install the `moku` Library and `mokucli` Utility

At a command prompt, install the `moku` library using `pip` . You can check that the installation succeeded by running the Python command listed below. If there is *no output* from the Python command, then the installation has succeeded. If you see an error message, refer to the [Troubleshooting](#) section below.

```
1 $: pip install moku
2 $: python -c 'import moku'
```

Install `mokucli` by downloading the installer from [Utilities](#) . You can easily check that the installation succeeded by running the command listed below from your command line. If the output begins the same as the output shown below, then the installation has succeeded. Read more about Moku CLI (`mokucli`) command line features [here](#).

```
1 $ mokucli --help
2 Usage: mokucli [OPTIONS] COMMAND [ARGS]...
3
4     MOKU command line utility
5
6     Version: X.X.X
7     ...
```

If the output does not match above, please refer to our [Knowledge Base](#) for troubleshooting.

### Note

It's important to make sure that the `mokucli` installation is in the same environment as, or is accessible by the `moku` package. Please refer to

### 3. Download the data files

The Moku Scripting API for Python requires data files to be downloaded before any program can be run. These data files may be several hundred megabytes. Please ensure you have a suitable internet connection before you proceed, this step is only required to be run whenever you install or upgrade the library.

You will need to point directly to your installation of the `moku` package. You can do this by finding the path of your moku installation, creating a data folder and downloading the data files to this path.

Find the path of your moku installation by looking at the location, in the example below this is the line: `Location: C:\Users\venv\Lib\site-packages` with `\moku\data` appended. You may need to manually create the data folder before downloading the bitstreams to this path using `mkdir`.

```
1      # finding the target path for moku python installation
2      $: pip show moku
3      Name: moku
4      Version: 3.3.3
5      Summary: Python scripting interface to the Liquid Instruments Moku hardware
6      Home-page: https://liquidinstruments.com
7      Author: Liquid Instruments
8      Author-email: info@liquidinstruments.com
9      License: MIT
10     Location: C:\Users\venv\Lib\site-packages
11     Requires: requests
12     Required-by:
13
14     # check for or create the data folder
15     $: mkdir C:\Users\venv\Lib\site-packages\moku\data
16
17     # download bitstreams for firmware version 600 to target from location
18     $: mokucli download 600 --target "C:\Users\venv\Lib\site-packages\moku\data"
19     Downloading latest instruments for firmware version 600...
20     [=====]
```

If you see an error message, refer to the [Troubleshooting](#) section below.

## 4. Find Your IP Address

For more detail and other options, see [Finding your IP Address](#)

```
1 $: mokucli list
2 Name           Serial HW    FW    IP
3 -----
4 MokuGo-000010  10   Go   600  192.168.###.###
```

## 5. Run Your First Program

Use your IP address as found above in place of the one below.

```
1 $ python
2 >>> from moku.instruments import Oscilloscope
3 >>> osc = Oscilloscope('192.168.###.###')
4 >>> osc.get_data()
```

If you are connected to your device via USB, ensure there are square brackets around the IP address, see [Specifying IPv6 Addresses](#) for more details.

```
1 $ python
2 >>> from moku.instruments import Oscilloscope
3 >>> osc = Oscilloscope('[fe80::7269:79ff:feb9:0000%0]')
4 >>> osc.get_data()
```

You should see an array of data captured from your newly-deployed Oscilloscope object. If not, please refer to the Troubleshooting below.

## Next Steps

Check out our [Python Examples](#) for more inspiration. Happy Coding!

## Troubleshooting

---

### 'python' is not recognized as an internal or external command, operable program or batch file

Python is not correctly installed on your Windows machine. Follow the instructions [here](#) .

### python: command not found

Python is not correctly installed on your Mac, or you're using a UNIX terminal on Windows that also cannot find Python. For Mac, follow the instructions [here](#) .

### The system cannot find the file

C:\Users\user\AppData\Local\Microsoft\WindowsApps\python.exe

Newer versions of Windows ship with Python installed by default. This error message indicates that the default installation has become corrupted somehow. Liquid Instruments recommends installing Python from scratch, according to the instructions [here](#) .

### ModuleNotFoundError: No module named 'pip'

You have tried to execute the `pip` command from within Python, rather than from a command prompt. Try using a terminal like Windows Terminal, Mac Terminal or `cmd.exe` .

### Listed Version is less than 3.5

You must upgrade your Python to 3.5 or newer. See the installation instructions for [Windows](#) or [Mac](#) .

### ModuleNotFoundError: No module named 'moku'

If you are trying to install the `moku` Library with `pip install moku` you may find that Python cannot find the newly-installed `moku` library. This commonly occurs for one of two reasons:

- The `pip` command didn't complete successfully, OR

If you are sure that the `pip` command completed successfully, then you can resolve the problem option by executing `pip` directly from the Python installation, rather than separately.

```
1 $: python -m pip install moku
2 $: python -c 'import moku'
```

### **ModuleNotFoundError: No module named 'zeroconf'**

In a released version of the `moku` package, `zeroconf` was not included in the required packages. Try running `pip install zeroconf` to remedy the issue.

### **Connection to (ip address) timed out, Max retries exceeded**

The Moku library was unable to connect to the Moku device. Ensure you have the correct IP address using the steps above.

If the IP address is correct, check that the network is reachable from your computer. The easiest way to achieve this is to simply try and connect using the Moku Desktop Application.

### **USB Connection Issues**

On Windows (only, not Mac or Linux), the Moku requires a driver to be installed in order to work over USB. This driver is installed automatically when you install the Moku Desktop Application. If you are working in an environment where you are not able to install the full Desktop Application, a standalone driver installer is available by contacting our [Support Engineers](#).

The Moku API uses IPv6 to connect to the Moku over USB. If your computer does not have IPv6 enabled (e.g. disabled by an Administrator) then the USB connection will not be operable. Please reach out to our [Support Engineers](#) for guidance and other options.

This extends to environments inside your computer where IPv6 may be limited, such as Microsoft Windows Subsystem for Linux v2 (WSL2). For information on possible workarounds for WSLv2 support, see the [original bug report](#) and [this resulting discussion](#).

## LocationParseError/InvalidURL: Failed to parse

This is usually seen when using USB and comes from the underlying libraries being unable to decipher the Link Local address used. This can be fixed by updating those libraries as follows.

```
1 $: python -m pip install --upgrade urllib3 requests
```

### Windows

On Windows, there is a small chance that the specific IPv6 address used by the Moku can trigger a bug in a core Python library [\[1\]](#), leading to an `InvalidURL` error. If the above steps don't help, the work-around is to replace the `%` character in the IP address with the sequence `%2525` (i.e. a doubly-escaped percent character). For example, if the original IP address ended in `023e%61` :

```
1 o = Oscilloscope('[fe80:0000:0000:0000:7269:79ff:feb9:023e%252561]')
```

## IPv6 (including USB) Connection Issues

There are some environmental limitations when using IPv6, including using the Moku USB interface. See [this section](#) for more information.

### Access Requests.session

The connection to the Moku device uses the Python `requests` module internally. To configure request session attributes, for example custom proxy or authentication parameters, prepend `session_` to the Session attribute and pass it to the instrument constructor. For example, `trust_env` can be configured as `session_trust_env` .

```
1 o = Oscilloscope("192.168.###.###", session_trust_env=False)
```

## Breaking changes with firmware version 600

# Important Notice: Breaking Changes in Firmware 600

Starting with firmware version 600, there are some breaking changes that require you to update your Python client package to version **3.3.1**. If you encounter the following error:

```
1 >>> i = Oscilloscope('192.168.###.###', force_connect=True)
2
3 Traceback (most recent call last):
4   File "<stdin>", line 1, in <module>
5   ...
6   File "moku/__init__.py", line 45, in __init__
7     if int(props['firmware']) < MIN_COMPAT_FW:
8         ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
ValueError: invalid literal for int() with base 10: '600.0'
```

This indicates that you are using an older version of the client package. To resolve this issue, please ensure you have the latest version installed.

## moku download Command [deprecated]

Previously you could download the bitstreams with the `moku` library with the following instructions. This has been deprecated and you should use the `mokucli` commands and instructions.

At a command prompt, install the `moku` library using `pip`. You can easily check that the installation succeeded by running the simple Python command listed below. If there is *no output* from the Python command, then the installation has succeeded. If you see an error message, refer to the [Troubleshooting](#) section.

```
1 $ pip install moku
2 $ moku download
3 $ python
4 >>> from moku.instruments import Oscilloscope
5 >>> osc = Oscilloscope('192.168.###.###')
6 >>> osc.get_data()
```

---

You should see an array of data captured from your newly-deployed Oscilloscope object. If not, please refer to the Troubleshooting below.

---

[← Overview](#)

[MATLAB](#)